Part1

Independent Variable:Sex(female & male)

dependent Variable: Total dollar value of R's stock

H0: The mean total dollar value of R's stock is the same for female and male respondents.

H1: The mean total dollar value of R's stock differs between female and male respondents.

Group Statistics

	Respondents sex	N	Mean	Std. Deviation	Std. Error Mean
Total dollar value of R's stock	MALE	97	415922.43	1151396.406	116906.593
	FEMALE	54	120765.24	437756.540	59571.120

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance		Mean	Std. Error	95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Difference	Difference	Lower	Upper
Total dollar value of R's stock	Equal variances assumed	8.202	.005	1.810	149	.036	.072	295157.187	163059.227	-27049.994	617364.368
	Equal variances not assumed			2.250	135.748	.013	.026	295157.187	131209.259	35678.584	554635.791

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Total dollar value of R's stock	Cohen's d	960372.360	.307	-.028	.641
	Hedges' correction	965240.484	.306	-.028	.638
	Glass's delta	437756.540	.674	.315	1.028

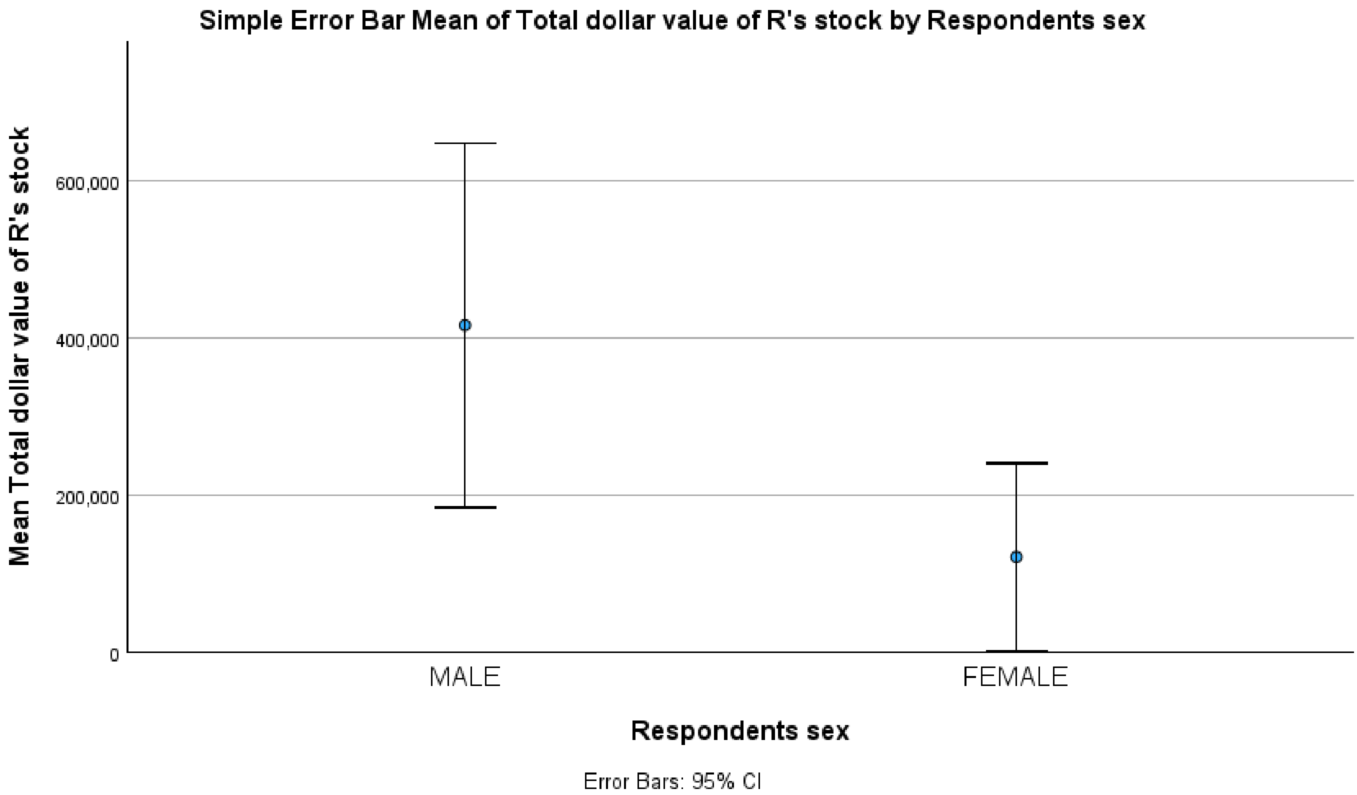
a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Result: Levene's test was significant (p =0.005< 0.05.Therefore, the "Equal variances not assumed" row was used for interpretation. The independent samples t-test showed a statistically significant difference in total dollar value of R's stock between female and male respondents. t =2.25, p =0.005. Male respondents (M = 415922.43) had a significantly higher mean total dollar value of R's stock than female respondents (M = 120765.24). Therefore, we reject the null hypothesis and conclude that gender group is significantly associated with differences in total dollar value of R's stock.

Test Assumptions:

An independent samples t-test was appropriate for this analysis because only two independent groups were compared (Male vs. Female) on a continuous dependent variable (Total dollar value of R's stock.). The dependent variable was assumed to be approximately normally distributed within each group. Finally, Levene's

Test for Equality of Variances was significant (p =0.005), indicating that the “Equal variances not assumed” results were used for interpretation.



The error bar chart visually displays the mean total dollar value of R's stock across gender groups with 95% confidence intervals. The male group has the higher mean dollaer value of stock than female group. The confidence intervals show relatively small overlap, suggesting meaningful differences in average dollar value of stock across groups. This visual pattern supports the independent samples t-test results, which indicated a statistically significant difference in mean total dollar value of R’s stock between gender groups (p =0.005).

Part 2

The purpose of this independent samples t-test exercise was to determine whether there is a significant difference in mean total dollar value of R’s stock between two categorical groups: male respondents and female respondents. The independent samples t-test is calculated by comparing the difference between the two group means relative to the variability within each group, using the formula for the t statistic, which incorporates the sample means, standard deviations, and sample sizes. Before interpreting the results, assumptions were assessed, including independence of groups, normality of the dependent variable, and homogeneity of variance using Levene’s Test. Levene’s Test was significant (p = 0.005), so the “equal variances not assumed” results were used. The test showed a statistically significant difference in mean dollar value of R’s stock between the two groups (p=0.005), leading us to reject the null hypothesis. Overall, the findings suggest that gender group membership is associated with differences in average total dollar value of R’s stock in this sample.